

## Reference part-2 (continuation)

| Sl. | Author       | Link                                                                                              | Abstract                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| 1   | Kumar et. al | <a href="https://doi.org/10.1016/j.est.2025.115464">https://doi.org/10.1016/j.est.2025.115464</a> | <p>The study focuses on the cooling of 18650 Lithium-ion <u>battery</u> pack exploiting <u>phase change material</u> and equally spaced fins surrounding the <u>battery</u>. The analysis includes an examination of heat transfer and temperature variation within/around the battery increasing the number of circular fins from 4 to 12, as well as varying the cell spacing from 8 mm to 12 mm. Furthermore, investigations are conducted for different <u>heat transfer coefficients</u> applied to the top surface of the battery pack, in the regime of both free and <u>forced convection</u>. Separate analyses are performed for <u>longitudinal fins</u>, and comparisons are drawn between circular and longitudinal fin configurations. Additionally, an experiment is undertaken for 4.4 W, 6.2 W &amp; 9.6 W mock-up battery for the validation of the present work. Analyses include temperature-time plots and contours of temperature and liquid fraction around the battery. Following this, the effectiveness of each configuration is determined in cooling the battery. The battery's temperature increases rapidly when exposed to air, reaching approximately 60 °C in just 400 s. However, with passive cooling using PCM, the time to reach the same temperature extends to 1800 s. The implementation of both fins and PCM aids in maintaining the battery's overall temperature at a lower level, thus mitigating the risk of thermal runaway-induced explosions. Using PCM and four circular fins increases the working time by 11.11 %. When eight circular fins are used, the working time (to reach 60 °C) is further extended by</p> |

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|   |                 |                                                                                                     | <p>10 %. Until 750 s, the longitudinal fin demonstrates more efficient heat transfer, resulting in a lower battery temperature compared to the circular fin. However, after 750 s, circular fins perform better in lowering the battery temperature. The working time increases by 25 % when the heat transfer coefficient of the medium changes from 5 W/m<sup>2</sup>K to 50 W/m<sup>2</sup>K. It is to be mentioned that the present work would be applicable for a battery pack either enclosed with adiabatic casing surfaces or with a greater number of cells where the impact of the casing of the battery pack on the interior batteries is neglected</p>                                                                                                                                               |
| 2 | Lanubile et. al | <a href="https://doi.org/10.1038/s44172-024-00304-2">https://doi.org/10.1038/s44172-024-00304-2</a> | <p>Accurate estimation of battery state of health is crucial for effective electric vehicle battery management. Here, we propose five health indicators that can be extracted online from real-world electric vehicle operation and develop a machine learning-based method to estimate the battery state of health. The proposed indicators provide physical insights into the energy and power fade of the battery and enable accurate capacity estimation even with partially missing data. Moreover, they can be computed for portions of the charging profile and real-world driving discharging conditions, facilitating real-time battery degradation estimation. The indicators are computed using experimental data from five cells aged under electric vehicle conditions, and a linear regression</p> |

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|    |           |                                                                                                 | <p>model is used to estimate the state of health. The results show that models trained with power autocorrelation and energy-based features achieve capacity estimation with maximum absolute percentage error within 1.5% to 2.5%.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 3. | Su et. al | <a href="https://doi.org/10.1109/TTE.2022.3204843">https://doi.org/10.1109/TTE.2022.3204843</a> | <p>The accurate estimation of state of health (SOH) for lithium-ion batteries is significant to improve the reliability and safety of batteries in operation. However, many existing studies on battery SOH estimation are conducted on the premise of large sizable labeled training data acquisition without considering the time cost and experimental cost. To solve such issues, this article proposes a novel capacity prediction method for SOH estimation based on the battery equivalent circuit model (ECM), deep learning, and transfer learning. First, an actual charge–discharge experiment is carried out, and a simulation of the corresponding cycling process is conducted for virtual data acquisition using the battery equivalent model. Second, a convolutional neural network (CNN)-based feature extraction network is selected by conducting a performance comparison. Then, a capacity estimation model consisting of a feature extraction network, regressor, and feature alignment metric calculation</p> |

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|   |             |                                                                                                         | modules is generated. Several transfer learning methods are chosen for feature alignment metric calculation. Finally, a capacity estimation performance comparison is done for the final selection of the feature alignment metric calculation methods. The results illustrate that the capacity prediction model established using virtual data and the generative adversarial network (GAN)-based transfer learning method has ideal prediction performance (with the 0.0941 of the maximum test error in all capacity estimation situation)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4 | Pang et. al | <a href="https://doi.org/10.1016/j.jechem.2022.11.036">https://doi.org/10.1016/j.jechem.2022.11.036</a> | Accurate insight into the heat generation rate (HGR) of lithium-ion batteries (LIBs) is one of key issues for battery management systems to formulate thermal safety warning strategies in advance. For this reason, this paper proposes a novel physics-informed neural network (PINN) approach for HGR estimation of LIBs under various driving conditions. Specifically, a single particle model with thermodynamics (SPMT) is first constructed for extracting the critical physical knowledge related with battery HGR. Subsequently, the surface concentrations of positive and negative electrodes in battery SPMT model are integrated into the bidirectional long short-term memory (BiLSTM) networks as physical information. And combined with other feature variables, a novel <u>PINN</u> approach to achieve HGR estimation of LIBs with higher accuracy is constituted. Additionally, some critical hyperparameters of <u>BiLSTM</u> used in <u>PINN</u> approach are determined through Bayesian optimization algorithm (BOA) |

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|   |             |                                                                                                                         | <p>and the results of BOA-based BiLSTM are compared with other traditional BiLSTM/LSTM networks. Eventually, combined with the HGR data generated from the validated virtual battery, it is proved that the proposed approach can well predict the battery HGR under the dynamic stress test (DST) and worldwide light vehicles test procedure (WLTP), the mean absolute error under DST is 0.542 kW/m<sup>3</sup>, and the root mean square error under WLTP is 1.428 kW/m<sup>3</sup> at 25 °C. Lastly, the investigation results of this paper also show a new perspective in the application of the <u>PINN</u> approach in battery HGR estimation.</p>                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5 | Shen et. al | <a href="https://doi.org/10.1016/j.applthermaleng.2025.126200">https://doi.org/10.1016/j.applthermaleng.2025.126200</a> | <p>The thermal issues of lithium-ion batteries significantly affect their performance, safety, and lifespan, and are one of the main obstacles hindering the further development of electric vehicles. Accurate tracking and prediction of the temperature distribution of lithium-ion batteries provide feedback for the Battery Management System, which is crucial for ensuring the safe and efficient operation of the <u>battery</u> system. At present, temperature estimation methods for battery systems mostly rely on physical models with numerous parameters, which fail to reflect the overall temperature distribution of large-format batteries. On the other hand, data-driven models for battery temperature require large amounts of training data and are prone to overfitting and other problems. These models lead to problems such as low accuracy in temperature estimation for battery systems, a high demand for training data, and poor model interpretability. To address these issues, this paper proposes a Physics-Informed</p> |

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|   |                    |                                                                                                   | <p>Neural Network model based on the thermal physical model of large-format batteries. By embedding the thermal-physical model information of the battery into the neural network, the model can accurately predict the evolution of the lithium-ion batteries temperature under various operating conditions. Experimental results show that under different constant current and dynamic conditions, the maximum Root Mean Square Error and Mean Absolute Error of the Physics-Informed Neural Network model do not exceed 0.6 °C. Compared with traditional methods, this approach significantly enhances the precision and interpretability of the model with smaller datasets, providing a more accurate basis for temperature distribution in battery management system.</p>                                                                              |
| 6 | Catenaro and Onori | <a href="https://doi.org/10.1016/j.dib.2021.106894">https://doi.org/10.1016/j.dib.2021.106894</a> | <p>In this paper, we report data from lithium battery cells from: Panasonic NCR-18650B (3350 mAh), LG Chem INR21700-M50 (4850 mAh) and A123 Systems ANR26650m1-B (2500 mAh). They own the same anode composition, graphite-based, and different cathode chemistry: lithium-nickel-cobalt aluminum-oxide (NCA), lithium-nickel-manganese-cobalt-oxide (NMC) and lithium-iron-phosphate (LFP), respectively. In this study, six cell samples were tested for each chemistry. The experiments consist in fully discharging the cells from 100% state-of-charge until the cell cutoff discharge voltage. The discharge is performed under controlled temperature conditions, namely 5 °C, 25 °C and 35 °C, and subjecting the battery cells to galvanostatic discharge rates ranging from C/20 to 5C, for NCA and NMC, and from C/20 to 20C, for LFP chemistry.</p> |

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|  |  |  | <p>The IncuMax IC-500R thermal chamber provides the reference temperature to the cell. The input current profiles are configured via the MITS Pro-software, and transmitted through the TCP/IP connection to the Arbin measurement system and the Arbin LBT21024. Voltage, current and cell surface temperature are measured on each cell and for each experiment to characterize the cells in terms of discharge capacity, discharge efficiency, thermal robustness, specific energy and specific power. A comprehensive analysis of the data is found in <a href="#">[1]</a>.</p> |
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